Real Business Cycle Model

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1 Basic RBC model: Social Planner's Problem

Preference:

$$U = E_0 \left[\sum_{t=0}^{\infty} \beta^t \frac{C_t^{1-\sigma} - 1}{1-\sigma} \right]$$

expectation given information at $t = 0$

Technology:

$$C_t + K_t = Z_t \boxed{K_{t-1}}^{\alpha} N_t^{1-\alpha} + (1-\delta) K_{t-1}$$

The amount of capital is predetermined in last period.

Produced goods & depreciated capital in last period used for consumption & new capital.

In current period t, agent decides on K_t.

Z_t is the TFP which follows the process:

$$\log Z_t = (1-\psi) \underbrace{\log \bar{Z}}_{\text{steady state TFP}} + \psi \log Z_{t-1} + \varepsilon_t$$

usually normalize $\bar{Z} = 1 \implies \log \bar{Z} = 0$

Endowment:

$$\underbrace{N_t = 1}_{\text{time/labor endowment}}, \qquad \underbrace{K_{-1} > 0}_{\text{initial capital stock}}$$

Information: decision made based on all information I_t up to t.
Social planner maximize the social utility by choosing C_t, K_t:

$$\max_{\{C_t, K_t\}} U = \max_{\{C_t, K_t\}} E_0 \left[\sum_{t=0}^{\infty} \beta^t \frac{C_t^{1-\sigma} - 1}{1-\sigma} \right]$$

subject to a budget constraint:

$$\forall t, \qquad C_t + K_t = Z_t K_{t-1}^{\alpha} N_t^{1-\alpha} + (1-\delta) K_{t-1}$$

$$N_t = 1, \qquad K_{-1} > 0$$

Set up dynamic Lagrangian equation:

$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \left\{ \frac{C_t^{1-\sigma} - 1}{1-\sigma} + \lambda_t \left[Z_t K_{t-1}^{\alpha} + (1-\delta) K_{t-1} - C_t - K_t \right] \right\}$$

F.O.C.:

$$\frac{\partial \mathcal{L}}{\partial C_t} = 0 \qquad \Rightarrow \qquad C_t^{-\sigma} - \lambda_t = 0 \tag{1}$$

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial K_t} = 0 \qquad & \Rightarrow \qquad -\lambda_t \beta^t + E_t\left[\lambda_{t+1}\beta^{t+1}\left(\alpha Z_{t+1} K_t^{\alpha-1} + 1 - \delta\right)\right] = 0 \\ & \Rightarrow \qquad -\lambda_t + \beta E_t\left[\lambda_{t+1}\left(\alpha Z_{t+1} K_t^{\alpha-1} + 1 - \delta\right)\right] = 0 \end{aligned} \tag{2}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_t} = 0 \qquad \Rightarrow \qquad C_t + K_t - Z_t K_{t-1}^{\alpha} - (1-\delta) K_{t-1} = 0$$

Transversality Condition:

$$\lim_{t \to \infty} \beta^t E_t \left[\lambda_t K_t\right] = 0$$

Plug in (1) into (2):

$$\beta E_t\left[C_{t+1}^{-\sigma}\left(\alpha Z_{t+1} K_t^{\alpha-1} + 1 - \delta\right)\right] = C_t^{-\sigma}$$

$$\Rightarrow \qquad 1 = \beta E_t\left[\left(\frac{C_t}{C_{t+1}}\right)^{\sigma}\left(\alpha Z_{t+1} K_t^{\alpha-1} + 1 - \delta\right)\right]$$

We have a dynamic system of (K_t, C_t, Z_{t+1})

$$K_t = Z_t K_{t-1}^{\alpha} + (1-\delta) K_{t-1} - C_t, \tag{1}$$

$$1 = \beta E_t \left[\left(\frac{C_t}{C_{t+1}} \right)^{\sigma} \left(1 - \delta + \alpha Z_{t+1} K_t^{\alpha-1} \right) \right], \tag{2}$$

$$\log Z_t = (1-\psi) \log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim iid\, N\left(0, \sigma_\varepsilon^2\right), \tag{3}$$

$$\text{state variables } (K_{t-1}, C_t), \qquad \text{shock variable } Z_t.$$

Three variables, three equations: #(equation) = #(variable)

2 What is the economic intuition behind Lagrangian multiplier?

Lagrangian multiplier is the shadow price of constrained resources. Marginal value of relaxing or tightening a constraint. For a problem,

$$V(b) = \max_x f(x) \qquad \text{subject to } g(x) \le b$$

$$\mathcal{L}(x, \lambda) = f(x) + \lambda \left(b - g(x) \right)$$

- If constraint is binding $g(x) = b$, λ shows how much the objective function would increase if you relax the constraint slightly.——shadow value

Envelope theorem:

$$\frac{dV(b)}{db} = \left. \frac{\partial \mathcal{L}(x, \lambda; b)}{\partial b} \right|_{x^*, \lambda^*} \qquad \Rightarrow \qquad V'(b) = \lambda^*(b)$$

$$V(b + \Delta) - V(b) \approx \lambda^*(b) \Delta$$

- If constraint is non-binding $g(x) < b$, $\lambda^* = 0 \qquad \Rightarrow \qquad V'(b) = 0$: A relaxation of constraint will not improve the objective function.

eg: RBC:

$$W_t = \max_{\{C_{t+j}, K_{t+j+1}\}_{j=0}^{\infty}} E_t \sum_{j=0}^{\infty} \beta^j \frac{C_{t+j}^{1-\sigma}}{1-\sigma}$$

subject to

$$C_{t+j} + K_{t+j+1} = F\left(Z_{t+j}, K_{t+j}\right) + (1-\delta) K_{t+j} + a_t$$

$$a_t \text{ appear only at } j = 0, \text{ perturb only one period's constraint}$$

$$\mathcal{L} = E_t \sum_{j=0}^{\infty} \beta^j \left[\frac{C_{t+j}^{1-\sigma}}{1-\sigma} + \lambda_{t+j} \left(F(\cdot) + (1-\delta) K_{t+j} - C_{t+j} - K_{t+j+1} + a_{t+j} \right) \right]$$

$$\frac{\partial W_t}{\partial a_t} = \left. \frac{\partial \mathcal{L}_t}{\partial a_t} \right|_{\text{at optimum}} = \lambda_t$$

$$\frac{\partial \mathcal{L}}{\partial C_t} = \beta^t C_t^{-\sigma} - \beta^t \lambda_t = 0 \qquad \Rightarrow \qquad C_t^{-\sigma} = \lambda_t$$

$$\Rightarrow \qquad \frac{\partial W_t}{\partial a_t} = \lambda_t = C_t^{-\sigma} = U_C\left(C_t\right) = \frac{\partial\left(\text{max attainable lifetime utility } W_t\right)}{\partial\left(\text{one more unit of resources at } t\right)}$$

What is transversality condition (TVC)?

Boundary condition at infinity that rules out Ponzi-scheme: leaving valuable resources unspent forever.

In a typical RBC model,

$$\text{TVC:} \qquad \lim_{T\to\infty} \beta^T U_C\left(C_T\right) K_{T+1} = 0$$

The discounted marginal utility value of the capital stock carried into the far future must vanish. Otherwise you're wasting utility by not consuming some of it earlier.

At large T, if we increase C_T by ε and decrease future capital by ε:

$$\text{First-order loss in capital:} \qquad \beta^{T+1}\lambda_{T+1}\varepsilon$$

$$\text{First-order gain in utility:} \qquad \beta^T U_C \varepsilon$$

$$\Rightarrow \qquad \lim_{T\to\infty} \beta^T U_C K_{T+1} = 0$$

in far future, you can not gain by this operation

3 Basic RBC model: competitive market approach

divide social planner into household and firm

*Household's problem: Household's own capital K_{t-1} & one unit of labor $N_t = 1$. They lend capital and labor to firms and earn interest R_t & wage W_t.

$$\max_{\{C_t,K_t,N_t\}_{t=0}^{\infty}} E_0\left[\sum_{t=0}^{\infty}\beta^t \frac{C_t^{1-\sigma}-1}{1-\sigma}\right]$$

s.t.

$$C_t + K_t^{(s)} = W_t N_t^{(s)} + R_t K_{t-1}^{(s)} + (1-\delta) K_{t-1}^{(s)}$$

(s) means supply

No-ponzi Scheme condition (TVC):

$$\lim_{t\to\infty} E_0\left[\left(\prod_{s=0}^{t} R_s^{-1}\right) K_t^{(s)}\right] = 0$$

*Firm's problem: Representative firm with tech Z_t, hire capital & labor to produce, and sells output to maximize profit. Given W_t, R_t, firm chooses $N_t^{(d)}$ & $K_{t-1}^{(d)}$.

$$\Pi_t = \max_{\left\{K_{t-1}^{(d)},N_t^{(d)}\right\}} Z_t \left(K_{t-1}^{(d)}\right)^{\alpha}\left(N_t^{(d)}\right)^{1-\alpha} - W_t N_t^{(d)} - R_t K_{t-1}^{(d)}$$

(d) means demand

where tech follows AR(1):

$$\log Z_t = (1-\psi)\log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim iid\, N\left(0, \sigma_\varepsilon^2\right)$$

Solve household's problem with Lagrangian Multiplier Method:

$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\sigma}-1}{1-\sigma} + \lambda_t \left(R_t K_{t-1}^{(s)} + W_t N_t^{(s)} + (1-\delta) K_{t-1}^{(s)} - C_t - K_t^{(s)} \right) \right]$$

$$\frac{\partial \mathcal{L}}{\partial C_t} = 0 \qquad \Rightarrow \qquad C_t^{-\sigma} - \lambda_t = 0 \tag{1}$$

$$\frac{\partial \mathcal{L}}{\partial K_t} = 0 \qquad \Rightarrow \qquad -\beta^t \lambda_t + \beta^{t+1} E_t \left[\lambda_{t+1} (R_{t+1} + 1 - \delta) \right] = 0$$

$$\Rightarrow \qquad \lambda_t = \beta E_t \left[\lambda_{t+1} (R_{t+1} + 1 - \delta) \right] \tag{2}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_t} = 0 \qquad \Rightarrow \qquad C_t + K_t^{(s)} = W_t N_t^{(s)} + R_t K_{t-1}^{(s)} + (1-\delta) K_{t-1}^{(s)} \tag{3}$$

Euler equation. Plug (1) into (2):

$$\underbrace{C_t^{-\sigma}}_{\text{MU of 1 unit } C \text{ today}} = \underbrace{\beta}_{\text{discount to today}} E_t \left[\underbrace{C_{t+1}^{-\sigma}}_{\text{MU of 1 unit } C \text{ tomorrow}} \underbrace{(R_{t+1} + 1 - \delta)}_{\text{return of extra unit of } K \text{ tomorrow}} \right]$$

LHS: The utility of 1 unit of consumption today.

RHS: The expected utility of sacrificing 1 unit of consumption today, instead save as capital, and carry over to next period, get R_{t+1} (rent earning) & get $(1-\delta)$ unit of depreciated capital, consume $R_{t+1} + 1 - \delta$ in the next period at marginal utility of consumption $\lambda_{t+1} = C_{t+1}^{-\sigma}$. Finally, discount to current value by multiplying β.

$$\text{LHS} = \text{RHS} \qquad \Rightarrow \qquad \text{Intertemporal trade-off ("no arbitrage")}$$

- A TFP shock on Z_t change the level through resource constraint, and dynamics through the Euler equation.

Resource constraint:

$$C_t + K_t = Z_t K_{t-1}^{\alpha} N_t^{1-\alpha} + (1-\delta) K_{t-1}$$

Solve firm's problem with Lagrangian multiplier method:

$$\frac{\partial}{\partial K_{t-1}^{(d)}} \left[Z_t \left(K_{t-1}^{(d)} \right)^{\alpha} \left(N_t^{(d)} \right)^{1-\alpha} - W_t N_t^{(d)} - R_t K_{t-1}^{(d)} \right] = 0$$

$$\Rightarrow \qquad \alpha Z_t \left(K_{t-1}^{(d)} \right)^{\alpha-1} \left(N_t^{(d)} \right)^{1-\alpha} = R_t \tag{4}$$

$$\text{MPK} = R_t$$

$$\frac{\partial \Pi_t}{\partial N_t^{(d)}} = 0 \qquad \Rightarrow \qquad (1-\alpha) Z_t \left(K_{t-1}^{(d)}\right)^{\alpha} \left(N_t^{(d)}\right)^{-\alpha} = W_t \tag{5}$$

$$\text{MPL} = W_t$$

Close the model: Market clearing conditions

Capital market:

$$K_{t-1}^{(d)} = K_{t-1}^{(s)} \qquad \cdots \quad \text{capital demand = supply}$$

Labor market:

$$N_t^{(d)} = N_t^{(s)} = 1 \qquad \cdots \quad \text{labor demand = supply}$$

Goods market:

$$C_t + K_t = Z_t K_{t-1}^{\alpha} N_t^{1-\alpha} + (1-\delta) K_{t-1} \qquad \cdots \quad \text{Resource constraint}$$

Walras Law: if there are m markets, then we need $m-1$ number of market clearing conditions. The last market is automatically cleared.

Thus, here we need two out of three. We keep resource constraint & capital market clearing by dropping (d) & (s). We also plug in constant labor supply, demand.

How resource constraint is derived?

Competitive market $\Rightarrow \Pi_t = 0$:

$$Z_t \left(K_{t-1}^{(d)}\right)^{\alpha} \left(N_t^{(d)}\right)^{1-\alpha} = W_t N_t^{(d)} + R_t K_{t-1}^{(d)}$$

(3) from household's problem:

$$C_t + K_t^{(s)} = W_t N_t^{(s)} + R_t K_{t-1}^{(s)} + (1-\delta) K_{t-1}^{(s)}$$

$$(s) = (d) \Rightarrow C_t + K_t = Z_t K_{t-1}^{\alpha} N_t^{1-\alpha} + (1-\delta) K_{t-1}$$

Produced goods + depreciated capital from last period is divided into use of capital and consumption

System of equations of the competitive market version of the model:

$$\text{(i)} \quad K_t = Z_t K_{t-1}^{\alpha} + (1-\delta) K_{t-1} - C_t \qquad \cdots \quad \text{resource constraint}$$

$$\text{(ii)} \quad 1 = E_t \left\{ \beta \left(\frac{C_t}{C_{t+1}} \right)^{\sigma} \left[\alpha Z_{t+1} K_t^{\alpha-1} + (1-\delta) \right] \right\}$$

$$\text{(iii)} \quad \log Z_t = (1-\psi) \log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim iid\, N\left(0, \sigma_\varepsilon^2\right)$$

* The competitive market version is equivalent to Social planner's version. (isomorphic)

$\Rightarrow$ Same equilibrium outcome

* The competitive market version of RBC model is a decentralized version of social planner RBC.

4 First and second welfare theorem in dynamic equilibrium

*First welfare theorem. Provided that all product and factor markets are competitive and there are no externalities in production or consumption, dynamic competitive equilibrium are Pareto Optimal.

"no friction" world in economics

*Second welfare theorem. All Pareto optimal allocation can be decentralized as a dynamic competitive equilibrium.
1^{st} Welfare theorem in RBC: CE allocation solves the social planner's problem $\Rightarrow$ Competitive equilibria are Pareto Optimal.

2^{nd} Welfare theorem in RBC: Social planner allocation solves the competitive equilibrium $\Rightarrow$ Pareto efficient allocation is supported by CE ($\exists$ competitive prices that support it).

Both theorems rely on:

1) perfect competition (price takers)

2) No externalities, no taxes/distortions, no market power.

3) Complete contracting:no missing markets (rep agent RBC avoids this)

4) Concave utility, convex technology set (CRS + concavity)

5) TVC / no-Ponzi to rule out pathological asset plans

Question 1. If there is a exogenous TFP shock Z_t, which moves more on impact C_t or K_t

$$Y_t = C_t + i_t \qquad \Rightarrow \qquad \Delta Y = \Delta C + \Delta i$$

$$Y_t = Z_t K_{t-1}^{\alpha} \qquad \Rightarrow \qquad Z_t \uparrow \Rightarrow Y_t \uparrow$$

$$\log Z_{t+1} = (1-\psi)\log \bar{Z} + \psi \log Z_t + \varepsilon_t \qquad \Rightarrow \qquad Z_t \uparrow \Rightarrow Z_{t+1} \uparrow \Rightarrow MPK_{t+1} \uparrow \Rightarrow R_{t+1} \uparrow$$

$$i_t = K_t - (1-\delta) K_{t-1} \qquad \Rightarrow \qquad K_t \uparrow \Rightarrow i_t \uparrow$$

$$1 = \beta E_t \left[\left(\frac{C_t}{C_{t+1}} \right)^{\sigma} (R_{t+1} + 1 - \delta) \right] \qquad \Rightarrow \qquad R_{t+1} \uparrow \Rightarrow \frac{C_t}{C_{t+1}} \downarrow \Rightarrow C_{t+1} \uparrow > C_t \uparrow$$

Thus, $\quad Z_t \uparrow \Rightarrow C_t \uparrow,\ C_{t+1} \uparrow\ \&\ i_t \uparrow$
K_{t-1} is pre-determined, thus at current period t, $\Delta K_{t-1} = 0$.

$$i_t \uparrow \gg C_t \uparrow ? \qquad \text{Ambiguous for now.}$$

1) but in data, it is true
2) 1° σ captures intertemporal substitution

$$\sigma \uparrow \qquad \Rightarrow \qquad \text{stronger consumption smoothing} \qquad \Rightarrow \qquad \underset{\text{more likely to hold}}{i_t \uparrow \gg C_t \uparrow}$$

2° ψ captures shock persistence

$$\psi \uparrow \qquad \Rightarrow \qquad i_t \uparrow \gg C_t \uparrow \qquad \text{more likely to hold}$$

3° α captures MPK

$$\alpha \uparrow \qquad \Rightarrow \qquad i_t \uparrow \gg C_t \uparrow \qquad \text{more likely}$$

4° δ captures depreciation

$$\delta \uparrow \qquad \Rightarrow \qquad i_t \uparrow \gg C_t \uparrow \qquad \text{less likely}$$

5° β captures patience

$$\beta \uparrow \qquad \Rightarrow \qquad i_t \uparrow \gg C_t \uparrow \qquad \text{more likely}$$

But in normal calibration, where

$$\sigma = 2, \qquad \psi = 0.95, \qquad \alpha = \frac{1}{3},$$

$$\delta = 0.025 \ (\text{quarter data}), \qquad \beta = 0.99 \ (\text{quarter data})$$

$$i_t \uparrow \gg C_t \uparrow \qquad \text{and} \qquad std(i_t) > std(C_t)$$

Question 2: What if households sell capital to firms and firms sell back to household after production?

Household's problem:

$$\max_{\{C_t,K_t,N_t\}_{t=0}^{\infty}} E_0\left[\sum_{t=0}^{\infty}\beta^t\frac{C_t^{1-\sigma}-1}{1-\sigma}\right]$$

s.t.

$$C_t+K_t^{(s)}=W_tN_t^{(s)}+R_tK_{t-1}^{(s)}$$

$$\Rightarrow \quad \begin{aligned} C_t^{-\sigma} &= \beta E_t\left[C_{t+1}^{-\sigma}R_{t+1}\right], \quad (1)\\ C_t+K_t^{(s)} &= W_tN_t^{(s)}+R_tK_{t-1}^{(s)}, \quad (2)\end{aligned}$$

Firm's problem:

$$\max_{\left\{K_{t-1}^{(d)},N_t^{(d)}\right\}}\Pi_t=\max\left\{Z_t\left(K_{t-1}^{(d)}\right)^{\alpha}\left(N_t^{(d)}\right)^{1-\alpha}-W_tN_t^{(d)}-R_tK_{t-1}^{(d)}+(1-\delta)K_{t-1}^{(d)}\right\}$$

$$\Rightarrow \quad \begin{aligned} \alpha Z_t\left(K_{t-1}^{(d)}\right)^{\alpha-1}\left(N_t^{(d)}\right)^{1-\alpha}+(1-\delta) &= R_t, \quad (3)\\ (1-\alpha)Z_t\left(K_{t-1}^{(d)}\right)^{\alpha}\left(N_t^{(d)}\right)^{-\alpha} &= W_t, \quad (4)\end{aligned}$$

System of equations is exactly the same as SP's problem. (isomorphic)
Here, R_t is no longer rental rate, but the gross return factor: rental + resale value.
Define the usual rental rate as

$$r_t=R_t-(1-\delta)$$

$$(3)\Rightarrow MPK=r_t$$

$$\Pi_t=Z_tK^{\alpha}N^{1-\alpha}+(1-\delta)K-RK-WN=Z_tK^{\alpha}N^{1-\alpha}-(R-(1-\delta))K-WN$$

$$=Z_tK^{\alpha}N^{1-\alpha}-rK-WN$$

$$(1)\Rightarrow \qquad C_t^{-\sigma}=\beta E_t\left[C_{t+1}^{-\sigma}(r_{t+1}+1-\delta)\right]$$

$$(2)\Rightarrow \qquad C_t+K_t=W_tN_t+r_tK_{t-1}+(1-\delta)K_{t-1}$$

$\Rightarrow$ Setting is totally isomorphic.

5 Supplementary notes

5.1 Technology-shock responses of C, I, and K

Under a positive technology shock, K_t does *not* move on impact, because it is predetermined. Thus, on impact, both C_t and I_t rise, while K_t remains unchanged.

The more subtle question is whether C_t or I_t rises more. In the model, this is not a pure theoretical certainty independent of parameters: it depends on the parameter values.

Empirically, however, the data typically show that consumption rises by less, while investment is more volatile.Later in the course, this can be seen more clearly from the impulse response functions.

5.2 Definitions of return

In the baseline setup, R denotes the *capital rent*, namely the rental price.

Later, in Q2, we define

$$r = R - (1 - \delta).$$

There, R is reinterpreted as the *gross return on capital*, i.e. the total return from holding one unit of capital across periods.

Under this notation:

$$R = r + (1 - \delta).$$

Hence:

- r corresponds to the baseline R, namely the rental price ;
- R in Q2 is the gross return on capital;
- the gross return consists of two parts:

$$\text{gross return} = \text{rental price} + \text{residual value of capital.}$$

6 Solve the RBC model

Find the Steady State (S.S.) of the economy, drop time subscript:

$$\bar{C} = \bar{Z}\bar{K}^{\alpha} + (1-\delta)\bar{K} - \bar{K} \qquad \cdots \quad \text{Resource constraint}$$

$$1 = \beta\left[\alpha\bar{Z}\bar{K}^{\alpha-1} + (1-\delta)\right] \qquad \cdots \quad \text{Euler equation}$$

Two variables, $(\bar{K}, \bar{C})$ given parameters $(\bar{Z}, \beta, \alpha, \delta)$

$$C_t = C_{t+1} = \bar{C}, \qquad K_t = k_{t+1} = \bar{K}, \qquad Z_t = Z_{t+1} = \bar{Z}, \qquad N_t = 1,\ \forall t$$

$$\bar{K} = \left[\frac{\alpha\bar{Z}}{\frac{1}{\beta} - (1-\delta)}\right]^{\frac{1}{1-\alpha}}, \qquad \bar{C} = \bar{Z}\bar{K}^{\alpha} - \delta\bar{K}, \qquad \bar{Y} = \bar{Z}\bar{K}^{\alpha}$$

$$\frac{\bar{C}}{\bar{K}} = \bar{Z}\bar{K}^{\alpha-1} - \delta = \frac{\bar{Y}}{\bar{K}} - \delta = \frac{\frac{1}{\beta} - (1-\delta)}{\alpha} - \delta = \frac{\frac{1}{\beta} - 1 + (1-\alpha)\delta}{\alpha}$$

7 Log-linearization

(i) Resource constraint:

$$K_t = Z_t K_{t-1}^{\alpha} + (1-\delta)K_{t-1} - C_t \qquad \Rightarrow \qquad \bar{K} = \bar{Z}\bar{K}^{\alpha} + (1-\delta)\bar{K} - \bar{C}$$

$$\bar{K}\left(1 + \hat{k}_t\right) = \bar{Z}(1+\hat{z}_t)\bar{K}^{\alpha}\left(1 + \hat{k}_{t-1}\right)^{\alpha} + (1-\delta)\bar{K}\left(1 + \hat{k}_{t-1}\right) - \bar{C}(1+\hat{c}_t)$$

Using

$$\left(1 + \hat{k}_{t-1}\right)^{\alpha} \approx 1 + \alpha\hat{k}_{t-1}$$

we get

$$\bar{K} + \bar{K}\hat{k}_t = \bar{Z}\bar{K}^{\alpha}\left(1 + \hat{z}_t + \alpha\hat{k}_{t-1}\right) + (1-\delta)\bar{K} + (1-\delta)\bar{K}\hat{k}_{t-1} - \bar{C} - \bar{C}\hat{c}_t$$

Subtract steady state:

$$\bar{K}\hat{k}_t = \alpha\bar{Z}\bar{K}^{\alpha}\hat{k}_{t-1} + \bar{Z}\bar{K}^{\alpha}\hat{z}_t + (1-\delta)\bar{K}\hat{k}_{t-1} - \bar{C}\hat{c}_t$$

$$\Rightarrow \qquad \hat{k}_t = \left(\alpha\frac{\bar{Y}}{\bar{K}} + 1 - \delta\right)\hat{k}_{t-1} + \frac{\bar{Y}}{\bar{K}}\hat{z}_t - \frac{\bar{C}}{\bar{K}}\hat{c}_t$$

$*$ next period change in k equals

1) change in K_{t-1}, that result in change in output $\left(MPK = \alpha\bar{Y}/\bar{K}\right)$ & depreciated capital $1-\delta$

2) change in tech z that result in change in output

3) change in consumption C that result in change in investment.

(ii) Euler equation:

$$1 = \beta E_t\left[\left(\frac{C_t}{C_{t+1}}\right)^{\sigma}\left(1 - \delta + \alpha Z_{t+1}K_t^{\alpha-1}\right)\right]$$

Define

$$R_{t+1} = 1 - \delta + \alpha Z_{t+1} K_t^{\alpha-1} \qquad \Rightarrow \qquad \bar{R} = \frac{1}{\beta}$$

from

$$1 = \beta E_t \left[\left(\frac{C_t}{C_{t+1}} \right)^{\sigma} R_{t+1} \right]$$

we get

$$1 = E_t \left[\left(\frac{C_t}{C_{t+1}} \right)^{\sigma} \frac{R_{t+1}}{\bar{R}} \right]$$

Let

$$\frac{C_t}{C_{t+1}} = \frac{\bar{C} e^{\hat{c}_t}}{\bar{C} e^{\hat{c}_{t+1}}} = e^{\hat{c}_t - \hat{c}_{t+1}}, \qquad \frac{R_{t+1}}{\bar{R}} = e^{\hat{r}_{t+1}}$$

Thus

$$1 = E_t \left[e^{\sigma(\hat{c}_t - \hat{c}_{t+1}) + \hat{r}_{t+1}} \right]$$

use $e^x \approx 1 + x$ for small x:

$$1 \approx E_t \left[1 + \sigma \left(\hat{c}_t - \hat{c}_{t+1} \right) + \hat{r}_{t+1} \right]$$

$$\Rightarrow \qquad \sigma \hat{c}_t = \sigma E_t \hat{c}_{t+1} - E_t \hat{r}_{t+1} \tag{1}$$

Now derive $\hat{r}_{t+1}$. Let

$$MPK_{t+1} = \alpha Z_{t+1} K_t^{\alpha-1} \qquad \Rightarrow \qquad \overline{MPK} = \alpha \bar{Z} \bar{K}^{\alpha-1}$$

$$\bar{R} = 1 - \delta + \overline{MPK}$$

Now approximate MPK_{t+1} around steady state:

$$\alpha Z_{t+1} K_t^{\alpha-1} \approx \alpha \bar{Z} \left(1 + \hat{z}_{t+1}\right) \bar{K}^{\alpha-1} \left(1 + \hat{k}_t\right)^{\alpha-1}$$

$$= \overline{MPK} \left[1 + \hat{z}_{t+1} + (\alpha - 1) \hat{k}_t \right]$$

$$where \, (1+x)^a \approx 1 + ax \qquad \Rightarrow \qquad \left(1 + \hat{k}_t\right)^{\alpha-1} \approx 1 + (\alpha - 1) \hat{k}_t$$

$$R_{t+1} \approx (1-\delta) + \overline{MPK} + \overline{MPK}\left[\hat{z}_{t+1} + (\alpha - 1)\hat{k}_t\right] = \bar{R} + \overline{MPK}\left[\hat{z}_{t+1} + (\alpha - 1)\hat{k}_t\right]$$

$$\hat{r}_{t+1} = \log\left(\frac{R_{t+1}}{\bar{R}}\right) \approx \frac{R_{t+1} - \bar{R}}{\bar{R}} = \frac{\overline{MPK}}{\bar{R}}\left[\hat{z}_{t+1} + (\alpha - 1)\hat{k}_t\right]$$

$$\frac{\overline{MPK}}{\bar{R}} = \frac{\bar{R} - (1-\delta)}{\bar{R}} = 1 - \frac{1-\delta}{\bar{R}} = 1 - (1-\delta)\beta$$

Let

$$\tilde{\delta} \equiv 1 - (1-\delta)\beta$$

share of gross return that comes from new output rather than keeping old capital.

Then

$$\hat{r}_{t+1} = \tilde{\delta}\left[\hat{z}_{t+1} - (1-\alpha)\hat{k}_t\right] \tag{2}$$

Plug (2) into (1):

$$\sigma\hat{c}_t = \sigma E_t\hat{c}_{t+1} - \tilde{\delta}E_t\hat{z}_{t+1} + \tilde{\delta}(1-\alpha)\hat{k}_t$$

$$E_t(\hat{c}_{t+1} - \hat{c}_t) = \frac{\tilde{\delta}}{\sigma}\underbrace{\left(E_t\hat{z}_{t+1} - (1-\alpha)\hat{k}_t\right)}_{\text{log deviation of next period's MPK}}$$

Expected growth in consumption is proportional to expected returns.

$$E_t(\hat{c}_{t+1} - \hat{c}_t) \propto E_t\widehat{MPK}_{t+1}$$

$MPK \uparrow \Rightarrow$ want more consumption tomorrow $\Rightarrow$ save/invest more today $\Rightarrow C$ grows

$\sigma \uparrow \Rightarrow$ stronger smoothing, IES $\downarrow$ / more curvature $\Rightarrow$ households hate changing C across time

$\tilde{\delta} \uparrow \Rightarrow$ returns move more with MPK $\uparrow \Rightarrow C$ responds by more

(iii) Technology process:

$$\log Z_{t+1} = (1-\psi)\log\bar{Z} + \psi\log Z_t + \varepsilon_t$$

$$\log\left(\bar{Z}e^{\hat{z}_{t+1}}\right) = (1-\psi)\log\bar{Z} + \psi\log\left(\bar{Z}e^{\hat{z}_t}\right) + \varepsilon_t$$

$$\Rightarrow \qquad \hat{z}_{t+1} = \psi\hat{z}_t + \varepsilon_t$$

the deviation of technology from S.S. is an AR(1) process

The log-linearized system of equation:

$$\hat{k}_t = \left(\alpha\frac{\bar{Y}}{\bar{K}} + 1 - \delta\right)\hat{k}_{t-1} + \frac{\bar{Y}}{\bar{K}}\hat{z}_t - \frac{\bar{C}}{\bar{K}}\hat{c}_t$$

$$\alpha\frac{\bar{Y}}{\bar{K}} + 1 - \delta = \overline{MPK} + 1 - \delta = \bar{R} = \frac{1}{\beta}$$

$$\frac{\bar{Y}}{\bar{K}} = \frac{\overline{MPK}}{\alpha} = \frac{\bar{R} - (1-\delta)}{\alpha} = \frac{\frac{1}{\beta} - (1-\delta)}{\alpha} = \frac{\tilde{\delta}}{\alpha\beta}$$

$$\frac{\bar{C}}{\bar{K}} = \bar{Z}\bar{K}^{\alpha-1} - \delta = \frac{\overline{MPK}}{\alpha} - \delta = \frac{\tilde{\delta}}{\alpha\beta} - \delta$$

$$\Rightarrow \qquad \begin{cases} \hat{k}_t = \frac{1}{\beta}\hat{k}_{t-1} + \frac{\tilde{\delta}}{\alpha\beta}\hat{z}_t - \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right)\hat{c}_t \\ \sigma E_t\hat{c}_{t+1} + \tilde{\delta}\left(1-\alpha\right)\hat{k}_t - \tilde{\delta}E_t\hat{z}_{t+1} = \sigma\hat{c}_t \\ \hat{z}_{t+1} = \psi\hat{z}_t + \varepsilon_t \end{cases}$$

$*$ Three equations, three variables $\left(\hat{k}_t, \hat{c}_{t+1}, \hat{z}_{t+1}\right)$

8 Solve the system in terms of exogenous parameters & pre-determined variables

Pre-determined: $\left(\hat{z}_t, \hat{k}_{t-1}\right)$

Write in matrix form:

$$\begin{pmatrix} 1 & 0 & 0 \\ \tilde{\delta}(1-\alpha) & \sigma & -\tilde{\delta} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \hat{k}_t \\ E_t \hat{c}_{t+1} \\ E_t \hat{z}_{t+1} \end{pmatrix} = \begin{pmatrix} \frac{1}{\beta} & \delta - \frac{\tilde{\delta}}{\alpha\beta} & \frac{\tilde{\delta}}{\alpha\beta} \\ 0 & \sigma & 0 \\ 0 & 0 & \psi \end{pmatrix} \begin{pmatrix} \hat{k}_{t-1} \\ \hat{c}_t \\ \hat{z}_t \end{pmatrix}$$

Because

$$\hat{z}_{t+1} = \psi \hat{z}_t + \varepsilon_t$$

is a stationary process, the dynamic property of system is pinned down by the deterministic part:

$\hat{z}_t$ is either 0 or converges to 0 since $0 < \psi < 1$

$$M\, E_t[x_t] = N x_{t-1}$$

where

$$x_{t-1} = \left[\hat{k}_{t-1} \quad \hat{c}_t\right]'$$

$$M = \begin{pmatrix} 1 & 0 \\ \tilde{\delta}(1-\alpha) & \sigma \end{pmatrix}, \qquad N = \begin{pmatrix} \frac{1}{\beta} & \delta - \frac{\tilde{\delta}}{\alpha\beta} \\ 0 & \sigma \end{pmatrix}$$

$$\det(M) = \sigma > 0 \qquad \Rightarrow \qquad M^{-1} \text{ exists}$$

Assume

$$M^{-1} = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$$

Impose $MM^{-1} = I$:

$$\begin{pmatrix} 1 & 0 \\ \tilde{\delta}(1-\alpha) & \sigma \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\Rightarrow \qquad b_{11} = 1, \quad b_{12} = 0, \quad b_{22} = \frac{1}{\sigma}, \quad b_{21} = -\frac{\tilde{\delta}(1-\alpha)}{\sigma}$$

$$\Rightarrow \qquad M^{-1} = \begin{pmatrix} 1 & 0 \\ -\frac{\tilde{\delta}(1-\alpha)}{\sigma} & \frac{1}{\sigma} \end{pmatrix}$$

$$\Rightarrow \qquad M^{-1}N = \begin{pmatrix} 1 & 0 \\ -\frac{\tilde{\delta}(1-\alpha)}{\sigma} & \frac{1}{\sigma} \end{pmatrix} \begin{pmatrix} \frac{1}{\beta} & \delta - \frac{\tilde{\delta}}{\alpha\beta} \\ 0 & \sigma \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{\beta} & \delta - \frac{\tilde{\delta}}{\alpha\beta} \\ -\frac{\tilde{\delta}(1-\alpha)}{\sigma\beta} & 1 + \frac{\tilde{\delta}(1-\alpha)}{\sigma}\left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right) \end{pmatrix}$$

Using

$$\delta - \frac{\tilde{\delta}}{\alpha\beta} = -\frac{\bar{C}}{\bar{K}},$$

we can write

$$M^{-1}N = \begin{pmatrix} \frac{1}{\beta} & -\frac{\bar{C}}{\bar{K}} \\ -\frac{\tilde{\delta}(1-\alpha)}{\sigma\beta} & 1 + \frac{\tilde{\delta}(1-\alpha)}{\sigma}\frac{\bar{C}}{\bar{K}} \end{pmatrix}$$

$$E_t[x_t] = M^{-1}N x_{t-1}$$

Jacobian matrix

Thus, $M^{-1}N$ shows the property of roots.

$$T = 1 + \frac{1}{\beta} + \frac{\tilde{\delta}(1-\alpha)}{\sigma}\frac{\bar{C}}{\bar{K}}$$

$$D = \frac{1}{\beta}\left[1 + \frac{\tilde{\delta}(1-\alpha)}{\sigma}\frac{\bar{C}}{\bar{K}}\right] - \frac{\tilde{\delta}(1-\alpha)}{\sigma\beta}\frac{\bar{C}}{\bar{K}} = \frac{1}{\beta}$$

$$p(-1) = 1 + D + T = 2 + \frac{2}{\beta} + \frac{\tilde{\delta}(1-\alpha)}{\sigma}\frac{\bar{C}}{\bar{K}} > 0$$

$$p(1) = 1 + D - T = -\frac{\tilde{\delta}(1-\alpha)}{\sigma}\frac{\bar{C}}{\bar{K}} < 0$$

$$\Rightarrow \qquad \begin{cases} p(-1) > 0 \\ p(1) < 0 \end{cases}$$

$$\Rightarrow \qquad 1 \text{ stable root in } (-1, 1), \qquad 1 \text{ unstable root in } (1, +\infty)$$

Since we have one predetermined variable $\hat{k}_{t-1}$ and one jump variable $\hat{c}_t$,

$$\Rightarrow \qquad \text{the system is a saddle path equilibrium}$$

(from Blanchard and Kahn conditions)

9 What is Jacobian Matrix?

Partial derivative that tells how a change in today's state affects tomorrow's variables.

eg:

$$x_t = f(x_{t-1}, z_t)$$

$$J = \left.\frac{\partial f}{\partial x}\right|_{(\bar{x}, \bar{z})}$$

(Notes transcribed by CMT.)